

ActInf Textbook Group ~ Cohort 1 ~ Meeting 15

(Chapter 6, part 1)

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all right greetings everyone it is september 9th 2022 it's the 15th meeting for cohort one and we are uh discussing chapter six we're in the first of two discussions on chapter six of the textbook we're going to go to the questions page and start there where we have one question that we explored a bit last week in our welcome back and then there's at least one question prepared but anyone is welcome to add more questions here while we turn first to this question what are the four steps in the recipe to construct an active inference model and can we make templates that facilitate people walking through some of these stages how much model building do people prefer to be engaged in in this section of the textbook group do people want to have one or more model that they're personally developing do they want those to be scaffolded in a shared page to help increment models along and see models different stages yep brock and anyone else i just it it's the same problem as like the first half of the textbook without math it's not it's not going to go well this half without modeling it's not it's it's going to be quite difficult i think so um i think we should just get uh that out of the way that it's yeah it's not about whether you want to or not or whatever if you if you if you want to just understand it at a cursory philosophical perhaps level then maybe not but if you actually want to understand it then yeah we are going to have to model something some way it doesn't have to be the most complex most precise one but yeah i i wonder if there are if we could develop like maybe a couple examples or simple um [Music] ones that people might like to model i don't know yeah those are all going to be purely agent-based but thanks i agree i wanted to frame it as a question but i agree with that perspective um and following along as we we um structure a few specific cases the rat and the teammates the icicating which are used in the textbook then some of the examples that are used recurrently in the literature like birdsong and a few others and for these many scripts exist and helping people who might not have the setup either by with some walkthroughs okay here's how you get octave running so you can do this matlab script or here's the standalone dem demo because this textbook really only gives the

pointer to matlab scripts and methods like when they say um that the standard schemes can be applied they mean a matlab script and that's exactly what the step-by-step guide model stream one is built around and chapter seven on modeling and discrete time is going to be akin to the step-by-step guide but it's not um step by step by step by step it's more like it takes two bigger steps um okay how is this four-step recipe for active inference modeling similar or different than approaches that people have seen for systems modeling from other frameworks for example in a reinforcement learning or cybernetic modeling so something more recent and computational or something more pre-formal or computational should this recipe be surprising to people coming from a certain background are there sub steps that are relevant to consider brock yes i don't have you know extensive like formal modeling experience but i just necessarily kind of most of the things that i've done for the last half decade like entail some form of this and i don't see how you could model anything without going through these um basic questions here like this is it's a bit i don't see how this is specific or exclusive or whatever to active inference it's a bit ambiguous to me how that yeah i mean besides specifically degenerate model i guess but which system are we modeling another way to ask that is what questions are we trying to answer or who i think lyle brought this up like who's the who are you making the model for sort of thing because the same like you were saying in the just earlier about the particular choice of how you're modeling it uh is just a choice on that system and that has that entails you know particular kind of hidden states marco blanket et cetera that um may be relevant or may not be depending on the questions you know what you're trying to model um but that that arises from those questions i guess is what i'm saying thank you ali uh well i just wanted to mention that um referring back to your previous question uh i also i'm currently developing a model and actually an emotion perceiving agent uh by pretty much closely following those four steps but i'm getting some helps in that regard and how to actually implement those four steps in uh in my uh particular situation but um on the topic of modeling uh well at a very basic level we can categorize different approaches to modeling um in terms of answering to two to three different questions uh the wett models the how models and the why models at least that's the kind of approach um mostly done in neuroscientific modeling so uh in my opinion uh this these four steps and this recipe can be applied mostly to the how and the why what models are mostly a descriptive models of phenomena so um i think my connection yeah you're about here what models are mostly mostly uh the descriptive models uh and uh but when we want to uh model an agent uh a self-evidencing agent which tries to um which behaves as if uh it infers something about

its environment uh we have moved beyond uh just a descriptive stage and we're dealing with the how and probably even the why questions there thanks when i hear why i always think of tim bergens four questions the four wise aristotle's four y's y is a subjective modelers pluralistic playground and it doesn't lend itself to one specific even type of answer let alone a framing or a specific parameter combination and i think that's a huge aspect of the move from like what are systems to how am i going to model the system which is a part of the turn that happens when we start to model is we dispense with absolutism all of a sudden these questions about what we're trying to do our preferences and affordances and constraints and limitations and everything come into play and we might make a portfolio of models that are shining light on different aspects of even the very same phenomena so it's kind of like an empirically grounded pluralism and pragmatism when we start to be engaged in real useful modeling and that is something that i feel that one learns on the field not necessarily bystanding so that's just to say that the modeling is really important even for philosophical reasons to say nothing of some of the things that are coming to play in chapter seven and beyond where it's like yeah the c vector has this many entries because why and how many rows and columns does this have but once we understand why the rows and the columns and the shapes of these different entities are the way they are then we just have to use this scaffold to think about certain systems consider their form at a very structural level like is it in discrete time or in continuous time and so on and then we might set up not just one generative model but again families of them in the future we hope and expect that active block friends will facilitate us doing parameter sweeps across families of cognitive models and generative models um but even in this boutique phase of model construction that we're in in current year one would still say okay i want to have one model where it's either acceptable temperature or too hot and i want to have one with a three-state model too cold just right too hot here's going to be a four stage model and just being able to specify that again instills pluralism through action about how we're modeling a system which is the philosophical point that it seems like sometimes isn't grasped by those who don't engage in the specific discussions around modeling a system okay where does figure 6.1 come into play a rehearsal of some of the figures we've seen earlier in the book and many many other places this is the particular partition the particle is the internal and the active sensory states so internal plus blanket states equals particular states those are the particles if we were looking at dust under the microscope like brownian motion was describing initially the particles were the dust particle and then the external states were like that which was not the dust particle partitioning the generative process and the generative model off from

each other we're dealing in bayesian mechanics with particular states with particles with things that are able to engage in sensory motor type loops this is just bringing in a few uh more pieces than that which is just to call attention to the direction and the existence of the arrows in the work how particular is the free energy principle of aguilara at all and in follow-up work there's discussion around which topologies on this particular partition facilitate or enable which kinds of formal claims to be made for example one might find it kind of interesting that sensory states have a bidirectional arrow with external states or might find it equally interesting or differently interesting or less interesting that also active states an arrow pointing back to internal states of course the questions that we have raised all along the way of like okay let's just say that we're gonna model a person's arm what are the active states what are the sensory states what does it mean that there's a bi-directional arrow connecting them is this the only topology for this particular partition could we have the around the clock model external states influencing sensory then internal then active and just going around the clock without any crosstalk in the blanket without any back arrows is that plausible does that change the formalism on one hand how could it not change the formalism when it's changing the structure of how variables are related to each other or maybe there are formalisms that are actually abstracted away from any topology choice because what is being described are the flows and the gradients on these variables but then what are those variables i think i mean that just highlights this need for for modeling but if we if we just really briefly go back to that uh one it's like the fact that there's this bi-directional arrow between external states and sensory states is um a bit confusing um in that again the language like well if it is acting on the environment isn't that an action state you know or something like this um in what way is it bidirectional so so could you not draw that as two you know one arrow going to the sensory state one arrow from the sensory state to back to the environment and then say what functionally does that entail um and then you're you're doing this particular selection of of how you're modeling right but i think you have to do that um to understand why that is bi-directional there right um in that particular framing of the generated process generate model there right which is still to me also slightly nebulous but uh i think yeah just i the bi-directionality of it between the action and sense states that kind of makes sense almost um that you sense your actions and your actions or response to your sins but that's not really what's being done there precisely either right so just explicating that is what yeah we need to separate and explicate what another really important question is like this action perception loop is appealed to early and often it's the first picture in the textbook before the high road in the low road in 1.2 is one one we're

talking about agents and they're interfacing with the niche and we're going to end up calling how we model asian degenerative model and we're going to end up calling the niche and the dynamics of a niche and how it's influenced as the generative process and we're going to call the interface between the generative model and generative process markov blanket and it's going to inherit this technical definition from base graphs and we're going to add these variables and we're going to connect them a certain way but we're talking about the feedback loop with the agent and the niche oh so what is this then are the observations the sensory states and this isn't just like a low-hanging fruit like well then why isn't it like you know why couldn't we call them i mean they are called as x and y interestingly in the continuous time formulation and i get it that they're being distinguished here because this is the continuous time pomdp and the um discrete time pomdp um but where do we see active states here we see policies so are active states along the branch influencing the b matrix how the hidden states change through time there may be a very simple um trail from this action perception loop framing the discrete and continuous time pomdps yeah i think just to me the simple way if you instead of thinking of this as a circular thing where you're literally going from one state to the next state to the way you might do in a normal state machine if you're thinking of no the active when you have an active state that however it is changing that is necessarily entailing change in the sensory external state and the internal states and and vice versa here with the sensory states like that if you if you perform some action of some kind there's some active change in that state like you're necessarily changing the internal hidden state right so like if you you know grab a glass of water or something like this like the what you're modeling now has changed slightly right and the sensors your internal states and are doing the same thing where you know for the glass of water or whatever was this you know object just sitting there doing nothing but now it's become a thing to to drink out of and that's changed in some sense the kind of external states that are being fed into your sensory states right like it it's changed the markov blanket that you're kind of selecting for or relating to and i mean that yeah the pomdp it's i mean i think that visibly they you know do the same sort of thing where if you have this observation you're necessarily creating a new internal you know or sense state that is going to change your policy selection so and that's going to change your your sense states returning to the recipe one question that i think will reduce our uncertainty about as we really do the modeling how do we account for residuals and understand which modeling residuals which is to say variability in data that are not explained by the which and what amounts of residuals are acceptable so if we're doing a linear regression and the only data we had on hand was height

and the only data that we wanted to predict were weight we could just say we've used all the data and whether the model fits um with an r value of 0.1 or 0.9 we know what the residual is of height on weight regression and here's a number that describes what fraction of the variability was described with this regression our value of one being like it's all perfectly on a line our value of zero being like it's just a total scatter plot we can say how much variability is explained we're done with the empirical data that we had so we can't explain any more when we're looking at empirical traces of behavior explain even a small fraction of behavior for some systems it might be the case that the generative models already must be quite complex in another scenario it might be the case that a very simple like looks to the left are are uh 90 of the time followed by looking to the right i mean on average isn't that true so then when we are evaluating models whether in block prints sweeping across models or just qualitatively or in a boutique way are we looking to reduce the residual to zero what traces of data are we looking to explain variability in and then if we're not going to be with empirical traces of behavior looking to reduce our unexplained variance what is going to be the criteria by which we do model selection on the aic and the bic which are information criteria balance variability explained across families of models with their degree of parameterization penalizing having more parameters rewarding better explanations so seeking to find a balance with models that are on the kind of pareto frontier of explanatory and simple that's great that's like one extra nuance on top of merely explaining more variability with a smaller residual it also adds in the penalty for having more parameters so that um in theory and in practice stops the modeler from just okay well now we're going to add in the temperature in this other part of the world because one percent of the variability just happened to be explained so now we're just developing these like totally spiraling looking for more and more data sources to explain a decreasing amount of variability that's left so it's good to pull back from that edge through the use of information criteria but at the core of the criterion is still the imperative to reduce the residual to have a model that fits within a model the way that we fine-tune it so that it fits data well but not overfitting and then across models we can see an analogous process of wanting to model that fits without for example overfitting or over-including parameters um so i think one question which might be added here is like what data do we have if we're doing a didactic model and we just want to sketch it out and run run a simulation and just purely generate data that might be useful in certain settings but the question of what behavioral data we have is very non-trivial if we're going to be approaching this as an empirical analysis problem brock just another question there um for us what uh what observations must we make or something

is the related what data do we have if you don't have the data that there's if there's not sufficient affordance information contained in the data to model the problem then do we have to model our action selection at a second or third order yeah exactly to find yeah to realize the right epistemic value so that was figure six one discrete and continuous shallow deep and hierarchical so just to address this which might come up shallow versus deep is describing how iterated model is through time within one type of time so you could have the one hour shallow where hours are units or you could have the 10 hours where it goes one two three four five six seven eight nine ten or you could have a hundred depth of a hundred with a time click of one or one could have a hierarchical model where there's 10 hours that make like a deca hour there's 10 clicks of the deca hour to reach a planning horizon of 100 time steps deep is describing one unit or counter of time that counter is depth is describing strict hierarchical nesting and so um they can both be used separately or together to describe events over longer and longer time horizons crucially this is in the context of language processing the duration of the word transcends that of any phoneme and the sentence transcends that of any word in the sequence if we assume paragraphs are made of multiple sentences paragraphs always transcend and encompass sentences so anyone else want to just add up a point i'm sure that you fellows are familiar with this distinction but just wanted to point it out since temporality in these and doing model selection on different forms of temporality is going to be an essential piece of the puzzle yeah and it also reminds me of the shankarian analysis theory that if you remember we talked about uh earlier uh and it's hierarchical i mean i mean it's temporal hierarchy that it tries to describe the whole musical piece uh as a kind of hierarchical layers which i think maps perfectly into this exam specific example of language parsing because that also is a kind of thing that should ideally be modeled in a very hierarchical way but the caveat here is unlike language at least in some languages are necessarily hierarchical music is at least tonal music or even non-tonal music are not they don't necessarily have this hierarchical structure so depending on the kind of music or the genre of music we might have uh either a very shallow and non-hierarchical structure or a very deep one so uh that's something that uh again refers back to uh the kind of system we're trying to model but even that kind of system needs to be more um i mean specified in a more granular level in order to have an effective modeling strategy there thanks for bringing in that domain like it makes me think of the intro the chorus the bridge the verse perhaps these are only in certain genre of music but analogously and then the measure transcends any beat in the measure with a special case of like a one one time signature or something and so on although language is described in a

strictly hierarchical sense i'm wondering how syntax and grammar actually create kind of a strange loop for example it's trivial to say that phonemes are nested within words and sentences and so on but what about the semantics of a sentence that contains commas and exceptions and then at the end the speaker says the previous sentence is not how you think it is is also causing kind of like a deep linguistic recall can be trivially modeled as containing nested units but will a model that narrowly considers the semantics of words than the semantics of clauses how will it be able to address some of these questions here i think we can point to a really excellent attribute of the active inference modeling framework in this discussion where the lw paper was cited we could just marvel at nested systems and draw blankets on blankets and so on and they can be drawn around every organelle and every little lipid granule in the neurons and or the brain regions and so we discussed a little earlier like doing structural model selection on which one of those are relevant however this does not mean we need to attempt to model the entire brain to develop meaningful pragmatic simulations of a single level for example if we wanted to focus on word processing we could address some aspects without having to deal with phoneme processes this means we can treat inputs from parts of the brain drawing inference about phonemes as providing observations from the perspective of word processing areas so one could say i'm going to make a figure 4.3 where s is the true word and phonemes are being passed as observations to my word inference engine and someone can say but aren't phonemes the result of inference and someone can say great make that module in block friends make that module and pass me that data and then we can do a nested model and we can graft those models together through the mark all blanket formalism not even the whole markov blanket bounding a specific particular entity but just the trivial markov blanket that exists insulating any two unconnected nodes in a base graph we can use that broader still technical concept of a markov blanket and just say all right phoneme observations come in and surely we could make a model where another type of information comes in and the inference is on phoneme and that's what's passed out so i think again to point to this advantage of active inference modeling it allows us to situate phenomena as complex multi-scale nested systems and also bite off what we can chew and makes it an empirical question of how grafting together which different structures might be useful but we don't need to um go all the way like turtles all the way down we can just say it's even if it were turtles all the way down we're modeling the third through the fifth turtle and the fifth turtle gets a top down prior and the third turtle gets sensory input from one below it and that's the part of the stack and this is the lateral width that we're modeling another advantage of active inference

which not even saying this is a unique advantage is that this goal directedness and corollary capacities like counterfactuals on goals and the way that counterfactuals on world states influence our goal selection nested goals and transient goals or conditional goals can be resolved in an unfolding action perception loop that's something that as this example of like wanting to enter an apartment of being goal driven over multiple time skills and then being able to again zoom in so it's like oh grab the keys okay well here's the key grabbing module and someone says well the key grabbing is actually um accomplished through this kind of grasping behavior then that could be modeled or you can just say we're subsuming finger grasping kinetics in the grasping module because the phenomena that we're trying to explain and the data we have is just about when things were grasped or maybe the only observations we have is when apartments were entered so we can actually i expect frame systems extremely expansively and clarify where we're doing formal modeling maybe this is just a philosophical or ontological framing of it but i'm i don't see how you can model anything well without admitting that it is either you know incomplete partially consistent or you know um but there is some hidden state which is probably not included in the and to me active inference kind of natively uh has an affordance at least at the very least a bookmark to do that and so whatever depth or hierarchy you model the system it's okay it's okay that there's still hidden state and it's incomplete um if you know if it sufficiently models answers the question at hand then you can always add another layer or you know make the model more complex that's just what we were going to do anyways nice comment i totally agree with that again thinking back to the linear model example where is that extra information about like um adjacent possibles for the model it's unstructured it's kind of like either the structured information is going to enter your linear regression just so and everything that's outside of your linear regression is um we're going to need to go back to the square one if we if we include a second observable we're going to have to go to square one and do the model selection all over again and test for all the repeat interactions again um but in these multi-scale models that we're discussing like you you said it was a bookmark it's kind of a nice way um frame it which is like the phonemes you're bookmarking or leaving an open usb port or something any thing that outputs a phoneme will plug into that part of the model anything is able to hear can be plugged in to the frog riveting anything that can see can be plugged into the visual component of the frog it's almost like consistent with all of this blanket talk we're being clear about what is modeled formally and the borders of the formal model are degrees of freedom as formulated to be expanded upon and composed one other i'm not sure the context uh it's something or ben ben gertz will say

attributed to um pinch right but um um that in circular mo um circular reasoning when your circle gets big enough it becomes coherent maybe another way of saying like that if your model is insufficient that if you add enough states enough you know if you expand your blanket enough eventually if you're modeling something correctly i guess uh you know it can the general process can can did match blanket states can can come into the line um some more active inference isms so exteroceptive proprioceptive and terrapeptive we're going to use active inference memory attention anticipation planning as inference that is what it looks like to integrate disparate fields using one model one family of models so you know others can work out some uh fun ways to communicate and frame it but i think when people are like like how is this simplifying when it might feel like it's bringing in a lot or even stepping on namespaces that people already have familiarity in like we discussed earlier you could ask how is memory related to attention or perhaps more saliently how will you model how memory is related to attention are you going to go on archive and just look for neural network architectures are you going to read a book written before computers were invented and look at prose are you going to dive into the psychoanalytic tradition or something that's maybe even clinically biomedical what's the move and active inference addresses that in these um uh theme of where's the people used that mean different things for four hundred we have bayes optimal behavior well then why do things go wrong and there's multiple layers to unpack first is going wrong is about your perspective on how the system perhaps how you prefer it to be or how you expected it to be but like given where you put the dish when it fell off the table that was just compatible with gravity so it was finding its optimal position and fragmentation where you placed it the ball rolled downhill as a consequence of where it was placed and the slope it was on framing behavior in that light we're modeling behavior as rolling to the bottom of a bowl what's the bowl what's the ball those are the questions but it's like chemical reactions proceeding under gibbs free energy minimization when we have policy selection driven by variational and expected free energy but how can things that go wrong be optimal it has to do with different parameters in the model given how they are the model performs optimally that doesn't mean computationally efficiently it doesn't mean adequately doesn't mean it satisfies but it may be framed as bayes optimal fixed and learned behaviors this is going to come into play in chapter seven there's an amazing continuity in theory and in simple examples with perception and learning which is to say that perception happens over faster time scales or perceptive like processes happen over faster time scales while learning like processes happen over slower time scales or even just more generally perception and learning refer to parametric

update processes and you might even be able to situate the same example both ways like if we see a ball move across our visual field are we perceiving the location of the ball are we inferring the location of the ball or are we updating and learning our position of the ball for those with a computational background learning often equates to parameter updating and so in that sense any kind of dynamical perception is learning for some backgrounds the difference between perception and learning will be seamless for other backgrounds those are going to sound like totally different processes i mean isn't perception when you see the book but then learning is like when you understand something about the book and then speaking from a more modeling perspective here a fixed model or fixing a parameter of a model reduces the model's complexity immensely or another way to say it taking a fixed parameter and making it learnable introduces multiple hyper parameters and there's no single way to address parameter learning you could say well we're taking we're doing moving average learning like a column filter or just a simple sliding window we're taking the average of the last three but now you have to parameter sweep across how many and with a single trial of data there may be um it may not be clear which learning strategy is implemented because especially if we want to differentiate learning strategies empirically we would want to see like multiple comparable trajectories within and amongst individuals in similar and different contexts but now you know our lab has gotten quite big and we're doing quite large model selections even for learning and updating parameters that might just conversationally seem really simple like preferences being updated so here's this description of when it comes to inference and perception as being fast changes and learning as slower changes though they're being modeled in a really analogous way it is worth noting in this book we exemplify rather simple generative models that are defined using tabular methods eg with explicit matrices or tensors for priors and likelihoods in small state spaces in comparison more sophisticated gms are being developed in machine learning deep learning robotics etc so is active inference deep learning good fellow generative adversarial networks with uh bengio and mirza but are we saying those are formally active inference systems or are we just saying that we can think of it like birdsong conceptually and then are we just adding a rapper or a descriptor to the gans that our non-active colleagues are building is it shining any new epistemic or pragmatic light on gans i mean here's a whole paragraph on gans home is behind oh that's chapter 10. it's like what happened but there's still a whole paragraph on did you see that how did that happen whoa i'm not sure what the optimal way to view a pdf is or that one exists but i don't think it's a browser it's not bad but it's not not optimal um yeah yeah i don't i would i interested in that you know

again the generative model oh i know what those are so general model like um and how you know specific specifically gans here and active inference like is is the rectangle a square uh sorry square rectangle rectangle not a square here is that you know what we're is being said or is there a degree of uh a um indexed blanket here of a varying degree um yep yep as far as i know and also you fellows might know like this is being pointed to variational auto encoders and variational bayesian inference are directly implicated as potentially recursive cortical networks are and world models if these bridges can be made solid like you said with squares and rectangles so that we can say all sum or none x are y or here's how you project this model to model it's going to just break a dam possibly certain applications of active any final comments on six and uh i think certainly by um two weeks from now let's uh look at the spm implementation and look at some of those that are pointed towards in the book as well as some of like the epistemic chaining and um pi mdp type models as well as anything that anyone else builds so any final comments uh yeah i just wanted to point out uh the recent work by uh feels at all especially uh one the two papers that person was also as a co-author uh is one of the their main objectives is to cross uh that bridge between active inference and bae uh via the ccd system so uh that bridge they're talking about in this book is already been investigated uh to be uh filled so yeah they're uh pretty interesting developments going uh in that regard okay any last notes before i stopped recording what is that paper i'll send you the link from the discord i forgot um maybe just one i don't know we haven't tried to make an effort on the code to like um lay down some basic examples from one you know the first half but um should there be some if we're gonna model stuff is should we have a subpage for that or do put something somewhere in the coda for that i don't know we've had this code section from the beginning yeah and just however it should be totally modifiable full control so however this and anything subpages need to be modified and the github link a textbook anything that people suggest just let us anything else